MMWAVE RadarSS Release Notes

1 Introduction

RadarSS firmware is responsible for configuring RF/analog and digital front-end in real-time. It also schedules periodic temperature based calibrations and monitors based on use need. This enables the mm-Wave front-end to be autonomous and capable of adapting itself to handle temperature and ageing effects, and to enable significant ease-of-use.

2 Release Overview

2.1 Platform and Device Support

The device and platforms supported with this release include:

Supported Devices	Release Status	Supported EVMs
xWR294xECO ES1.0	Production Release	xWR294xECO EVM
xWR2E4xECO ES1.0	Production Release	xWR2E4xECO EVM
xWR294xP ES1.0	Production Release	xWR294xP EVM
xWR2E4xP ES1.0	Production Release	xWR2E4xP EVM
xWR2x4xLC ES1.0	Production Release	xWR2x4xP EVM
xWR2x4xLC ES1.0	Production Release	xWR2x4xP EVM

NOTE: All xWR2x4xECO API's listed in mmWave-Radar-Interface-Control.pdf are applicable for xWR2x4xLC device

2.2 Release version

Version	Type
ROM: 2.5.4.0 PATCH: 2.6.3.1	Binary patch (7.55KB)

3 Release Contents

3.1 xWR2x4xP ES1.0 Patch contents

xWR2x4xP ES1.0 has RadarSS Firmware contents in ROM and the below DFP releases provide patch on top of ROM. The below DFP release are applicable only for ES1.0 through firmware patch updates.

3.1.1 DFP 2.4.15.0 Features and Enhancements

- xWR2x4xP is TIs 77GHz RF CMOS Radar SOC. Features supported in this firmware release:
    - xWR2x4xP RadarSS ROM Firmware and patch Firmware is derived from xWR294x/xWR254x DFP2.4 release baseline. It is backward compatible to DFP 2.4 supported APIs with minor updates for enhancement.
    - Supports 4 TX and 4 Real-only RX chains.
    - Synthesizer RF frequency supported 76 – 81GHz
        - VCO1: 76 – 77GHz
        - VCO2: 76 – 81GHz
    - Supports 7 functional profiles.
    - Supports 266MHz/us max slope
    - Supports programmable filter
    - Support for inter-chirp jitter mitigation
    - Synth frequency monitor (Live Mode) support for multiple profiles
    - Support for enabling 1 ns resolution for ADC start time in advanced chirp configuration
    - Supports FRC DCC clock monitor
    - Improved minimum chirp cycle time for advanced chirp configuration by allowing enable/disable control for parameters in advanced chirp configuration
    - Support to enable/disable OSCCLKOUTETH to provide reference clock for external ethernet PHY
    - Support for 20Mhz IF bandwidth
    - Improved TX Power
    - Improved RX Noise Figure

- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR2x4xP ES1.0 API details.

3.1.2 DFP 2.4.16.0 Features and Enhancements

- Updates to RF Gain Target and Max RX Gain
- Improved TX Power accuracy

3.1.3 DFP 2.4.17.0 Features and Enhancements

- Improved TX Phase shifter accuracy
- Improved TX Power accuracy

3.1.4 DFP 2.4.18.0 Features and Enhancements

- Fixes for Group1 ECC Aggregator SEC and DED error

3.1.5 DFP 2.4.18.1 Features and Enhancements

- No changes to patch binary from DFP 2.4.18.0
- Refer 3.2.5 in Feature/Change List, 4.1.1 in Unsupported features and APIs and 5 Known Issues for documentation updates

3.2 Feature/Changes List

3.2.1 Changes in DFP 2.4.15.0 release (with respect to xWR2944/xWR2544)

Item type	Key	Description
Feature	MMWAVE_DFP-2076	Updates for TX Power improvement
Feature	MMWAVE_DFP-2077	Updates for RX Gain accuracy

3.2.2 Changes in DFP 2.4.16.0 release (with respect to DFP 2.4.15.0)

Item type	Key	Description
Feature	MMWAVE_DFP-2012	Updated RF Gain target from 32,34, 36 dB to 34, 36, 38 dB
Feature	MMWAVE_DFP-2110	Updates for TX Power accuracy

Feature	MMWAVE_DFP-2112	Update to support max RX Gain of 46dB

3.2.3 Changes in DFP 2.4.17.0 release (with respect to DFP 2.4.16.0)

Item type	Key	Description
Feature	MMWAVE_DFP-2126	Updates for TX Phase shifter accuracy
Feature	MMWAVE_DFP-2115 MMWAVE_DFP-2128 MMWAVE_DFP-2151	Updates for TX Power accuracy

3.2.4 Changes in DFP 2.4.18.0 release (with respect to DFP 2.4.17.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2169	Fix to generate ECC_AGC_DED_ERROR when a single bit error occurs in BSS Static RAM. Device now enters safe state when ECC_AGG_SEC_ERROR occurs due to a single bit error in BSS Static RAM or a double bit error ECC_AGG_DED_ERROR occurs in BSS Mailbox / BSS Static RAM. Refer device Errata for more details
Bug	MMWAVE_DFP-2290	Fix to clear ESM Group1 ECC aggregator SEC error. When the ESM Group 1 error ECC_AGG_SEC_ERROR occurs for the first time, single bit error is corrected and async event AWR_AE_RF_ADV_ESMFAULT_STATUS_SB is generated. Subsequent single bit error event would correct the single bit error but async event would not get generated. This has been fixed now
Bug	MMWAVE_DFP-2268	Improved temperature sensor accuracy

3.2.5 Changes in DFP 2.4.18.1 release (with respect to DFP 2.4.18.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2395	Defeature “Rx FE disabled”, value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB. Added 20MHz IFA loopback frequency in IF_LOOPBACK_FREQ field of AWR_LOOPBACK_BURST_CONF_SET_SB.

4 Unsupported features and APIs (applicable to all DFP releases)

The following APIs and features are not validated fully at system level, it is recommended not to use these APIs in this and all previous DFP releases. This list of unsupported features is in addition to the list mentioned in known issues.

4.1 Unsupported Features

4.1.1 Functional APIs

API	Feature	Description
PCR self-test feature in AWR MONITOR_RF_DIG_LATENTFAULT_CONF_SB	PCR self-test	The PCR self-test is not supported in xWR2x4xP.
AWR_INTERCHIRP_BLOCKCONTROLS_SB	Inter-chirp power saving timing configurations	This API is not validated at system level. It is recommended not to use the same.
AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB	RX noise figure monitor	The RX noise figure monitor is not supported in xWR2x4xP.
AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB	Rx mixer input power monitor	The RX mixer input power monitor is not supported in xWR2x4xP.
TXn POWER_BACKOFF fields in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB	Calibration and Monitoring Frequency and TX Power limits API	Recommended to perform factory calibration with 0dB back-off set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API and, store and restore the calibration data from non-volatile memory. With this option, user can set nonzero back-off option in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API during normal runs to back-off monitor/calibration chirp's power.
AWR_MONITOR_VMON_CONF_SB	VMON configuration	This feature has not been validated at a system level. It is not recommended to use this.
SYNTH_VCO1_SLOPE and SYNTH_VCO2_SLOPE fields in the PLL_CONTROL_ VOLTAGE_VALUES in AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB	VCOx slope monitoring	This feature is not supported in xWR2x4xP.
LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX0, LPF_CUTOFF_STOPBAND_ATTEN_VALUE, LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX fields in **AWR_MONITOR_RX_IFSTAGE_RE**	LPF monitoring results ASYNC report	These fields are only supported for debug purposes and should not be used for production use-cases.

PORT_AE_SB ASYCNC report		
LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH fields in **AWR_MONITOR_RX_IFSTAGE_CONF_SB** API	LPF monitoring threshold fields in the IF STAGE monitoring configuration API	These fields are only supported for debug purposes and should not be used for production use-cases.
Some combinations of RX_GAIN and RF_GAIN_TARGET in PF_RX_GAIN field of the **AWR_PROFILE_CONF_SB** and **AWR_CONT_STREAMING_MODE_CONF_SET_SB** APIs.	(RX Gain, RF Gain Target) (44dB, 32dB) (46dB, 32dB) (46dB, 34dB)	Since the maximum IF gain is +10dB in xWR2x4x, the specified pairs of (RX Gain, RF Gain Target) settings are not supported.
Some values of the GLOBAL_RESET_MODE field of the **AWR_ADVANCE_CHIRP_CONF_SB** API are not supported when using Advanced Frame Configuration.	Value 0 - “End of frame”	When using advanced frame configuration in conjunction with advanced chirp configuration, the “end of frame” reset mode is not supported in advanced chirp config API.
b27 - Bus Safety test in field RF_POWERUP_BIST_STATUS_FLAGS of **AWR_RF_BOOTUPBIST_STATUS_GET_SB** API	Bus Safety test Status	This feature is not supported in xWR2x4xP.
b30 - BUS_SAFETY_PCR_ERROR b31 - BUS_SAFETY_BSS_SLV_ERROR in field ESM_GROUP1_ERRORS_LSB and b0 - BUS_SAFETY_BSS_MST_ERROR b1 - BUS_SAFETY_STATIC_MEM_ERROR b2 - BUS_SAFETY_MBOX_ERROR b3 - BUS_SAFETY_SEC_AGGREGATED_ERROR in field ESM_GROUP1_ERRORS_MSB of **AWR_RF_ADV_ESMFAULT_STATUS_GET_SB** API b30 - BUS_SAFETY_PCR_ERROR b31 - BUS_SAFETY_BSS_SLV_ERROR in field ESM_GROUP1_ERRORS_LSB and	Bus Safety ESM Error	This feature is not supported in xWR2x4xP.

b0 - BUS_SAFETY_BSS_MST_ERROR b1 - BUS_SAFETY_STATIC_MEM_ERROR b2 - BUS_SAFETY_MBOX_ERROR b3 - BUS_SAFETY_SEC_AGGREGATED_ERROR in field ESM_GROUP1_ERRORS_MSB of **AWR_AE_RF_ADV_ESMFAULT_SB** API		
b24 – FFT test in field RF_POWERUP_BIST_STATUS_FLAGS of AWR_RF_BOOTUPBIST_STATUS_GET_SB API b24 – FFT test in field DIG_MONITORING_ENABLES of **AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB** API b24 – FFT test in field DIG_MON_LATENT_FAULT_STATUS of AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB API	FFT test	This feature is not supported in xWR2x4xP.
b4 - DFE_PARITY_AND_OUT_ERROR in field ESM_GROUP1_ERRORS_LSB of **AWR_RF_ADV_ESMFAULT_STATUS_GET_SB** API b4 - DFE_PARITY_AND_OUT_ERROR in field ESM_GROUP1_ERRORS_LSB of **AWR_AE_RF_ADV_ESMFAULT_SB** API	DFE AND Parity ESM Error	This feature is not supported in xWR2x4xP.

b0 - Digital RX gain compensation enable in field DIGITAL_COMP_EN and DIGITAL_RX_GAIN_COMP field of **AWR_DIGITAL_COMP_EST_CONTROL_SB** API	Digital RX gain compensation	This feature is not supported in xWR2x4xP.
OSCCLKOUT_DIS in CASCADING_ PINOUTCFG field of the AWR_CHAN_CONF_SET_SB API	OSCCLKOUT	OSCLKOUT is not supported. It is recommended to keep OSCLKOUT disabled.
RX FE disabled mode. Value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB	RX FE disabled mode in loopback burst configuration API	RX FE disabled option in not supported for loopback burst configuration API

4.1.2 Debug APIs

API	**Feature**	**Description**
AWR_RF_PALOOPBACK_CFG_SB AWR_RF_LOLOOPBACK_CFG_SB AWR_RF_IFLOOPBACK_CFG_SB	Loopback enables	PA, LO and IF loopback APIs are not supported in functional mode, recommended to use only for debug.
AWR_RF_TEST_SOURCE_CONFIG_SET_SB AWR_RF_TEST_SOURCE_ENABLE_SET_SB	Test source feature	Test source feature is not supported in functional mode, recommended to use only for debug.
AWR_CONT_STREAMING_MODE_CONF_SET_SB AWR_CONT_STREAMING_MODE_EN_SB	Continuous streaming mode	Continuous streaming mode is not supported in functional mode, recommended to use only for debug.

5 Known Issues

Key	Severity	Description
MMWAVE_DFP-1649	Minor	**Noise metric indicators vary with temperature.** RF loopback-based TX and RX gain phase mismatch monitors include a noise metric indicator to enable detection of corruption by external interference. The noise metric varies with temperature. Strategy to determine thresholds for detecting interference is mentioned in a monitoring application note. The strategy includes temperature dependent thresholds.
MMWAVE_DFP-2105	Major	**ECC_AGC_DED_ERROR mapped to ESM_GROUP1_ERRORS_LSB** Double bit ECC errors in BSS static RAM and BSS mailbox indicated by ECC_AGC_DED_ERROR is mapped to ESM_GROUP1_ERRORS_LSB. ESM_GROUP1_ERRORS are

		group of non-fatal errors, however double bit ECC error ECC_AGC_DED_ERROR is a fatal error. **Workaround:** **It is recommended to treat** ECC_AGC_DED_ERROR in field ESM_GROUP1_ERRORS_LSB of AWR_AE_RF_ADV_ESMFAULT_STATUS_SB and AWR_RF_ADV_ESMFAULT_STATUS_SB should be treated as a fatal error
MMWAVE_DFP-2395	Major	RX FE Disabled mode, value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB when used to measure noise floor show random jump in noise floor **Workaround:** It is recommended to use IF loopback mode with an out of band tone frequency to measure noise floor instead of RX FE Disabled mode